

SPRING DATA JPA - COMPLETE DEEP LEARNING GUIDE

1. Core JPA Fundamentals

- JPA Specification vs Implementation
- ORM (Object Relational Mapping)
- Hibernate as Provider
- Entity Lifecycle: Transient, Persistent, Detached, Removed

2. Entity Mapping

- @Entity, @Table
- Primary Keys (@Id, strategies)
- @Column
- Embedded (@Embeddable, @Embedded)
- Enums (@Enumerated)
- Large Objects (@Lob)

3. Relationships

- One-to-One
- One-to-Many
- Many-to-One
- Many-to-Many
- Owning vs Inverse side
- Join strategies
- Bidirectional vs Unidirectional

4. Fetching & Performance

- LAZY vs EAGER
- Fetch Joins
- Entity Graphs
- N+1 Problem

5. Persistence Context

- First-level cache
- Entity identity

6. Dirty Checking

- Automatic updates

7. EntityManager

- persist, merge, remove, find, flush, clear

8. Transactions

- Boundaries, propagation, isolation

9. Repositories

- JpaRepository, CrudRepository

10. Query Mechanisms

- Method Queries
- JPQL
- Native Queries

11. Pagination & Sorting

- Pageable, Page, Slice

12. Projections

- DTO & Interface projections

13. Cascading

- ALL, PERSIST, MERGE, REMOVE

14. Orphan Removal

15. Locking

- Optimistic & Pessimistic

16. Caching

- First & Second-level cache

17. Performance Tuning

- Query optimization, batch fetching

18. Auditing

19. DTO vs Entity

20. Exception Handling

21. Bulk Operations

22. Advanced Topics

- Specifications
- Criteria API
- Soft delete

FINAL MINDSET:

Focus on performance, transactions, and efficient data handling.